



American
Urological
Association



CRTP
CLINICAL RESEARCH
TRAINING PROGRAM



Duke University
School of Medicine

Methods in Clinical Urology Research (MCURe) Workshop

Reporting Effect Size and Precision in Clinical Research

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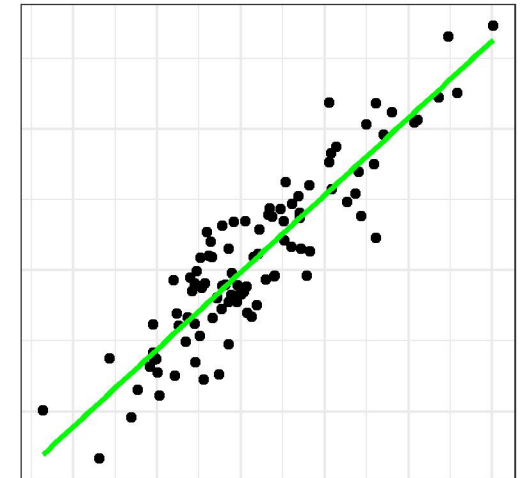
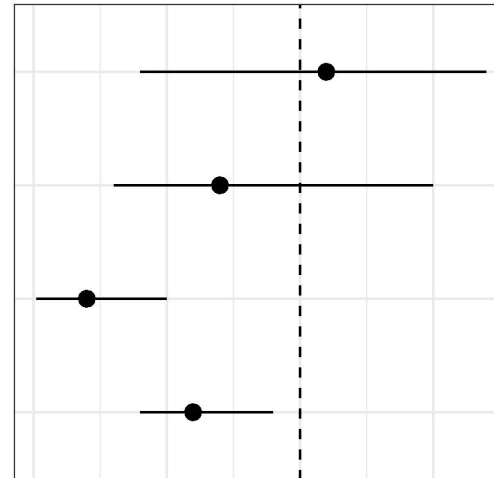
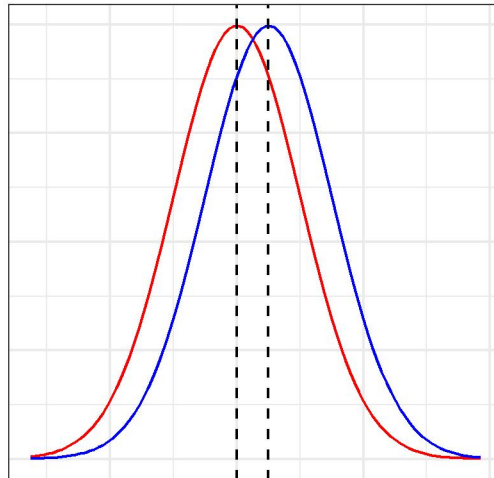
Topics covered today

1. Basics of effect size and precision
2. Reporting descriptive statistics
3. Reporting estimates for the main study questions
4. Reporting continuous versus categorical variables
5. Reporting time-to-event endpoints
6. Applying appropriate levels of precision

Basics of effect size and precision

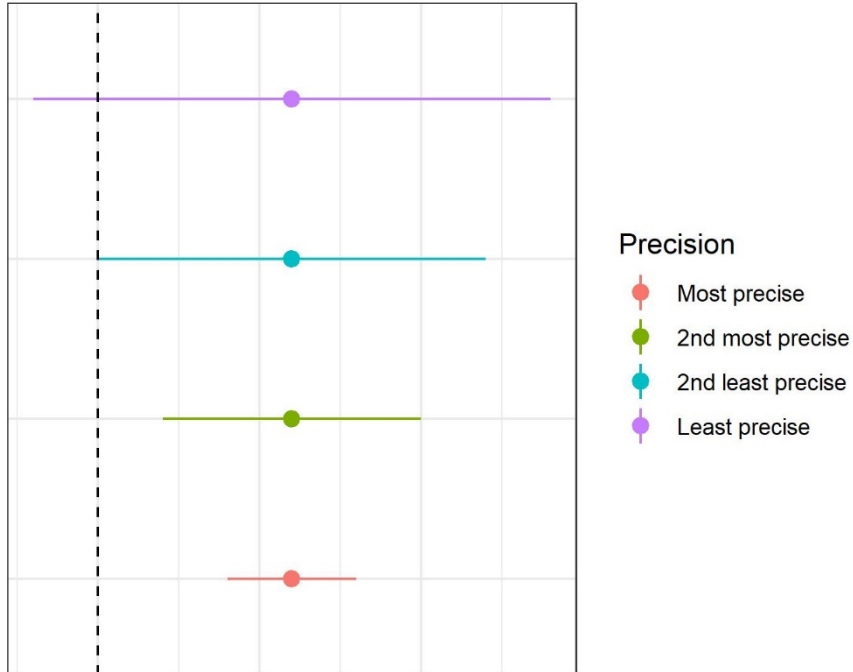
What is an effect size?

- A measure of the strength of association
- Examples:
 - Mean difference
 - Regression coefficient
 - Odds ratio
 - Hazard ratio
 - Correlation



What is precision?

A measure of statistical variability



The number of digits that contain
real information about a measure

123456.1234

Precision = number of significant digits

Reporting descriptive statistics

The anatomy of a “Table 1”

Characteristic	Overall, N = 193	No response, N = 132	Response, N = 61	p-value
Age, mean (SD)	47 (14)	46 (14)	50 (14)	0.091
Marker Level (ng/mL) , mean (SD)	0.91 (0.86)	0.84 (0.82)	1.07 (0.94)	0.10
Grade				>0.9
I	67 (35%)	46 (35%)	21 (34%)	
II	63 (33%)	44 (33%)	19 (31%)	
III	63 (33%)	42 (32%)	21 (34%)	
Sex				<0.001
Female	73 (38%)	34 (26%)	39 (64%)	
Male	120 (62%)	98 (74%)	22 (36%)	

Tip #1: avoid redundant statistics

Characteristic	Overall, N = 193	No response, N = 132	Response, N = 61	p-value
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Sex				<0.001
Female	73 (38%)	34 (26%)	39 (64%)	
Male	120 (62%)	98 (74%)	22 (36%)	

- Can obtain this information from the other two columns, not part of the presented statistical comparison

Tip #1: avoid redundant statistics

Characteristic	Overall, N = 193	No response, N = 132	Response, N = 61	p-value
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Sex				<0.001
Female	73 (38%)	34 (26%)	39 (64%)	
Male	120 (62%)	98 (74%)	22 (36%)	

- Can calculate the second row using simple arithmetic based on the column totals and the first row

Tip #1: avoid redundant statistics

Characteristic	No response, N = 132	Response, N = 61	p-value
Age, mean (SD)	46 (14)	50 (14)	0.091
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Tip #2: median/IQR preferred over mean/SD

Characteristic	No response, N = 132	Response, N = 61	p-value
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Grade			>0.9
I	46 (35%)	21 (34%)	
II	44 (33%)	19 (31%)	
III	42 (32%)	21 (34%)	
Female	34 (26%)	39 (64%)	<0.001

- Most continuous variables are not truly normally distributed
- Range only gives values for two patients

Tip #2: median/IQR preferred over mean/SD

Characteristic	No response, N = 132	Response, N = 61	p-value
Age, median (Q1, Q3)	46 (36, 55)	49 (43, 59)	0.091
Marker Level (ng/mL), median (Q1, Q3)	0.59 (0.20, 1.24)	0.98 (0.31, 1.53)	0.10
Grade			>0.9
I	46 (35%)	21 (34%)	
II	44 (33%)	19 (31%)	
III	42 (32%)	21 (34%)	
Female	34 (26%)	39 (64%)	<0.001

Tip #3: include unknown/missing numbers

Characteristic	No response, N = 132	Response, N = 61	p-value
Age, median (Q1, Q3)	46 (36, 55)	49 (43, 59)	0.091
Unknown	7	3	
Marker Level (ng/mL), median (Q1, Q3)	0.59 (0.20, 1.24)	0.98 (0.31, 1.53)	0.10
Unknown	6	4	
Grade			>0.9
I	46 (35%)	21 (34%)	
II	44 (33%)	19 (31%)	
III	42 (32%)	21 (34%)	
Female	34 (26%)	39 (64%)	<0.001

- Missing data reporting is essential to provide a full description of the study cohort (Note: not included in calculation of statistical tests)

Reporting estimates for the main study questions

Tip #1: report one estimate per question

Example:

A study seeks to address the following two questions:

1. Does biomarker level differ according to treatment group?
2. Is biomarker level associated with treatment response?

Does biomarker level differ according to treatment group?

Characteristic	Drug A, N = 98	Drug B, N = 102	Difference
Marker Level (ng/mL), mean (SD)	1.02 (0.89)	0.82 (0.83)	0.20
Unknown	6	4	

- 0.20 is the estimated mean difference in marker level between Drug A and Drug B

Is biomarker level associated with treatment response?

Characteristic	Odds ratio
Marker Level (ng/mL)	1.35

- A one-unit increase in marker level is associated with a 1.35-fold increase in the odds of response

Side notes on odds ratios:

- The odds of an event happening is the *probability that the event occurs* divided by the *probability that the event does not occur*
- The odds ratio (OR) is *the ratio of two odds* (e.g. the odds in males over the odds in females, or the odds in those with marker level $x+1$ over the odds of those with marker level x)
- In general, an odds ratio is *not* a risk ratio or relative risk, which is based on the ratio of two probabilities
- When the *prevalence of an event is very low* (i.e. rare events) the odds ratio can approximate the risk ratio

Tip #2: include confidence intervals

Question 1: Does biomarker level differ according to treatment group?

Characteristic	Drug A, N = 98	Drug B, N = 102	Difference	95% CI
Marker Level (ng/mL), mean (SD)	1.02 (0.89)	0.82 (0.83)	0.20	-0.05, 0.44
Unknown	6	4		

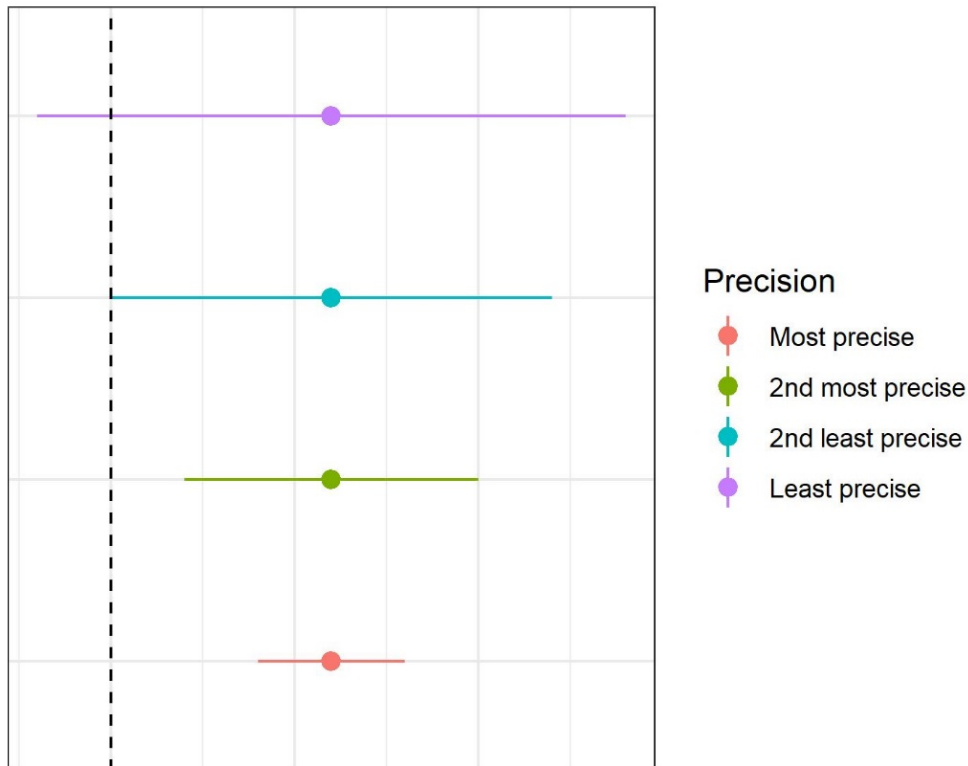
Question 2: Is biomarker level associated with treatment response?

Characteristic	Odds ratio	95% CI
Marker Level (ng/mL)	1.35	0.94, 1.93

- For Question 1, confidence interval includes 0 so we conclude there is no significant difference
- For Question 2, confidence interval includes 1 so we conclude there is no significant association

Interpreting statistical precision

- A confidence interval reflects the precision of the estimate



Wider confidence interval = less precise

More narrow confidence interval = more precise

Also reflects the **sample size**, e.g.:

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$



Reporting continuous versus categorical variables

Tip #1: do not treat categorical variables as continuous

Example:

A study seeks to examine whether tumor grade is associated with overall survival.

Characteristic	HR	95% CI	p-value
Grade	1.30	1.04, 1.63	0.023

Characteristic	HR	95% CI	p-value
Grade			0.075
I	—	—	
II	1.28	0.80, 2.05	
III	1.69	1.07, 2.66	

- Continuous assumes the difference between I and II, and II and III is the same – not usually the case

Tip #2: avoid categorizing continuous variables

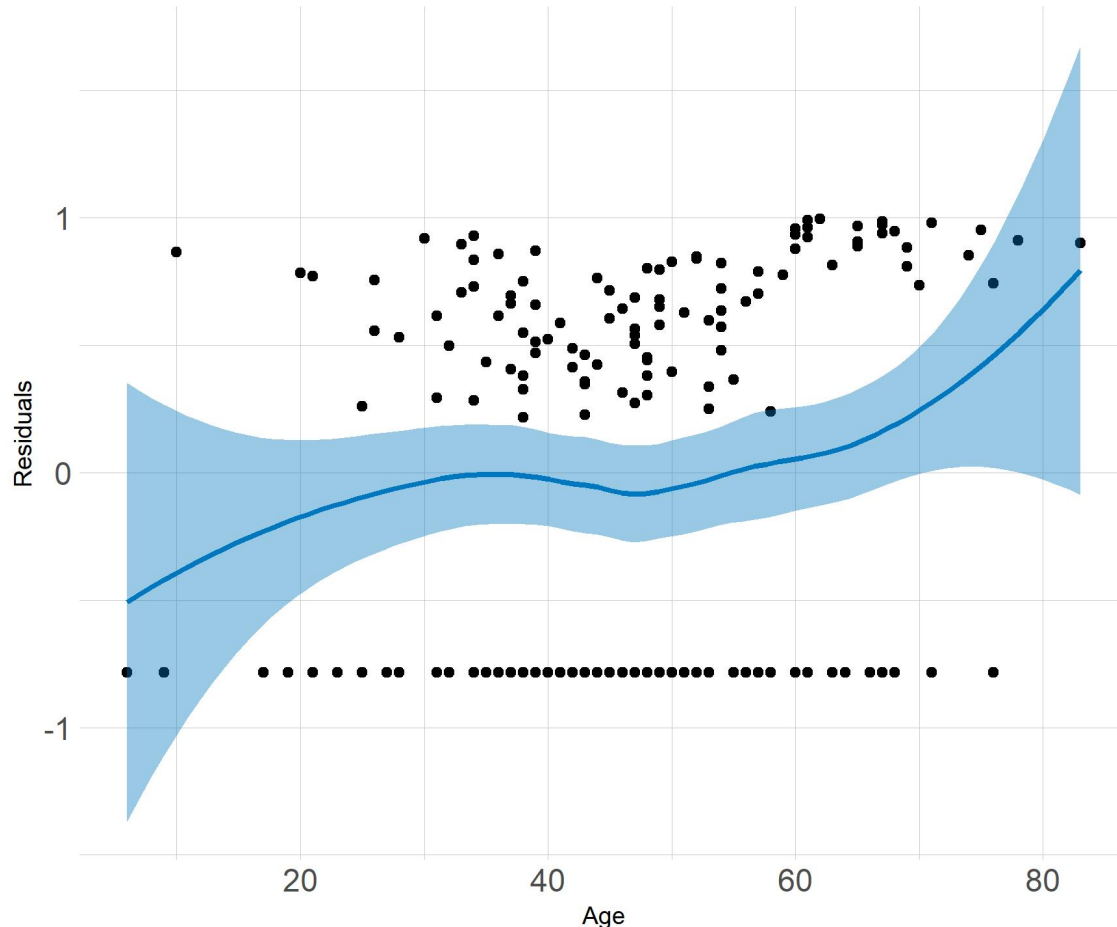
Example:

A study seeks to examine whether age is associated with overall survival.

Characteristic	HR	95% CI	p-value
Age categories			0.028
<60	—	—	
≥60	1.72	1.09, 2.72	

- Age ≥ 60 is associated with 1.72 times the hazard of death as compared to age < 60
- Implies no difference between 61 and 90 year old, but big difference between 61 and 59 year old

Tip #3: use graphical approaches to examine functional forms



Many approaches, for example:

- Simple scatter plot with regression line
- Plot of predicted value by continuous predictor with regression line
 - Possibly from nonlinear model
- Plot of residuals by continuous predictor with loess smooth

Combining tip #2 and tip #3

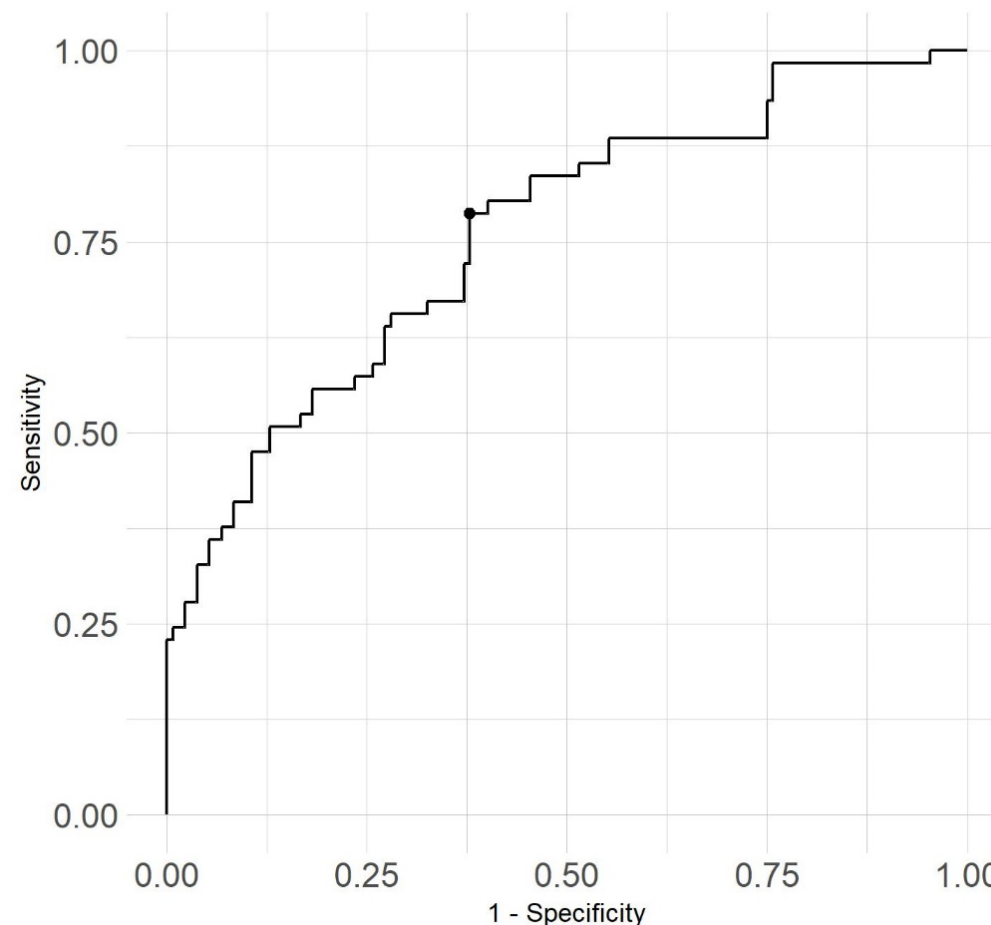
- Present age as a linear predictor (continuous) in the model for overall survival

Characteristic	HR	95% CI	p-value
Age	1.02	1.00, 1.03	0.042

- A one-unit increase in age is associated with a 1.02-fold increase in the hazard of death

Tip #4: avoid using statistical methods to obtain cutpoints

- **Example:** predict treatment response using marker value
- “Optimal” cutpoint according to Youden’s index is 0.4692
- But what is the clinical meaning of this?
- Are sensitivity and specificity truly of equal importance?



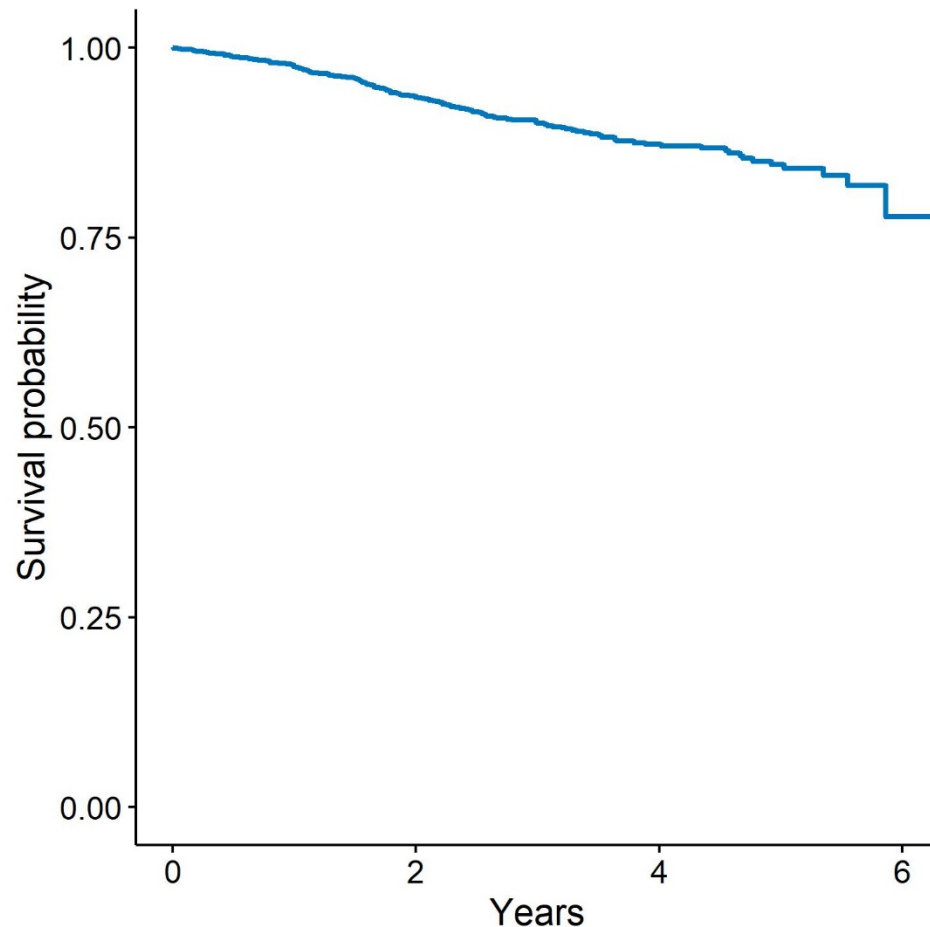


Reporting time-to-event endpoints

Tips for reporting time-to-event data

1. Report the number of events but not the proportion
 - Do: “138 patients died from any cause”
 - Don’t: “138 patients (9.2%) died from any cause”
2. Report median follow-up among the event free
 - Do: “Median follow-up among survivors was 2.7 years (IQR: 1.5-4.)”
 - Don’t: “Median follow-up was 2.6 years (IQR: 1.4-3.9)”
3. Describe start and end of follow-up and censoring details
 - Do: “Patients were followed from the date of diagnosis until the date of death. Patients still alive were censored at their date of last contact”
4. Use landmark analysis or time-dependent covariates if the status of a variable is not known at time zero (i.e. adjuvant therapy and time zero is date of surgery)

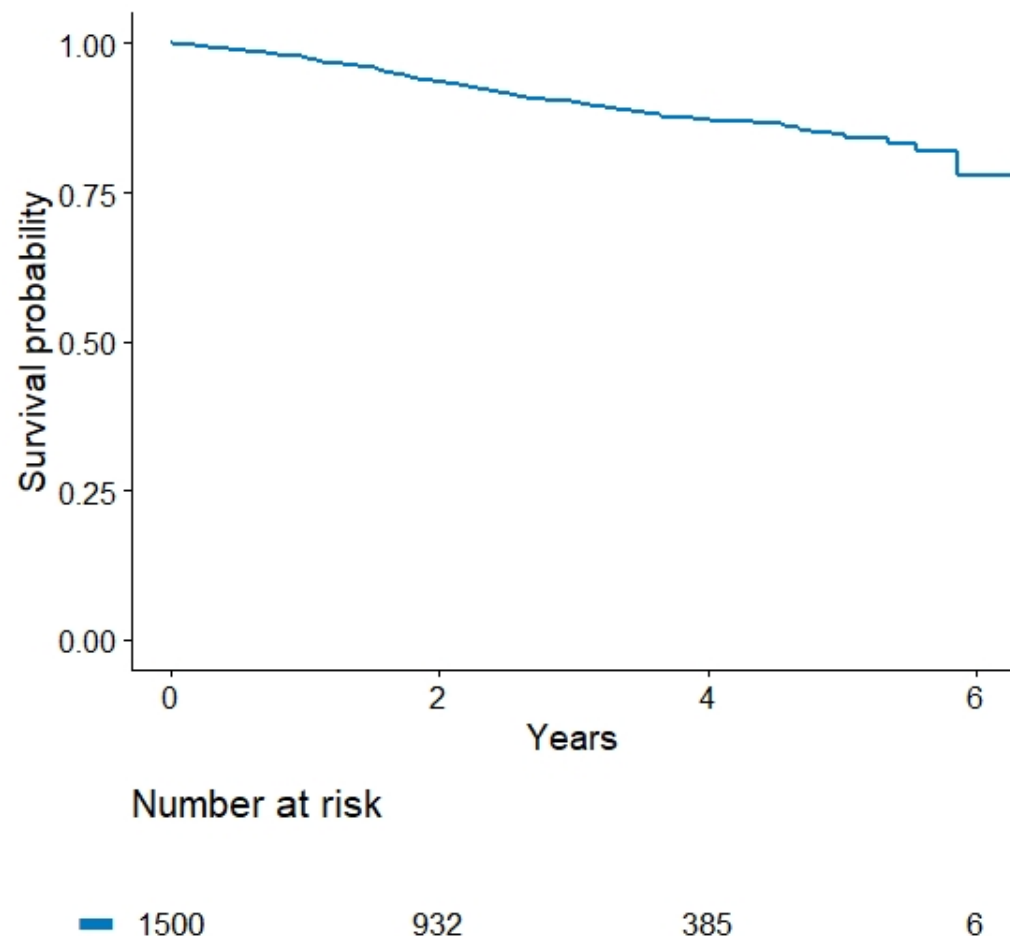
Tip #5: Include number at risk on K-M plots



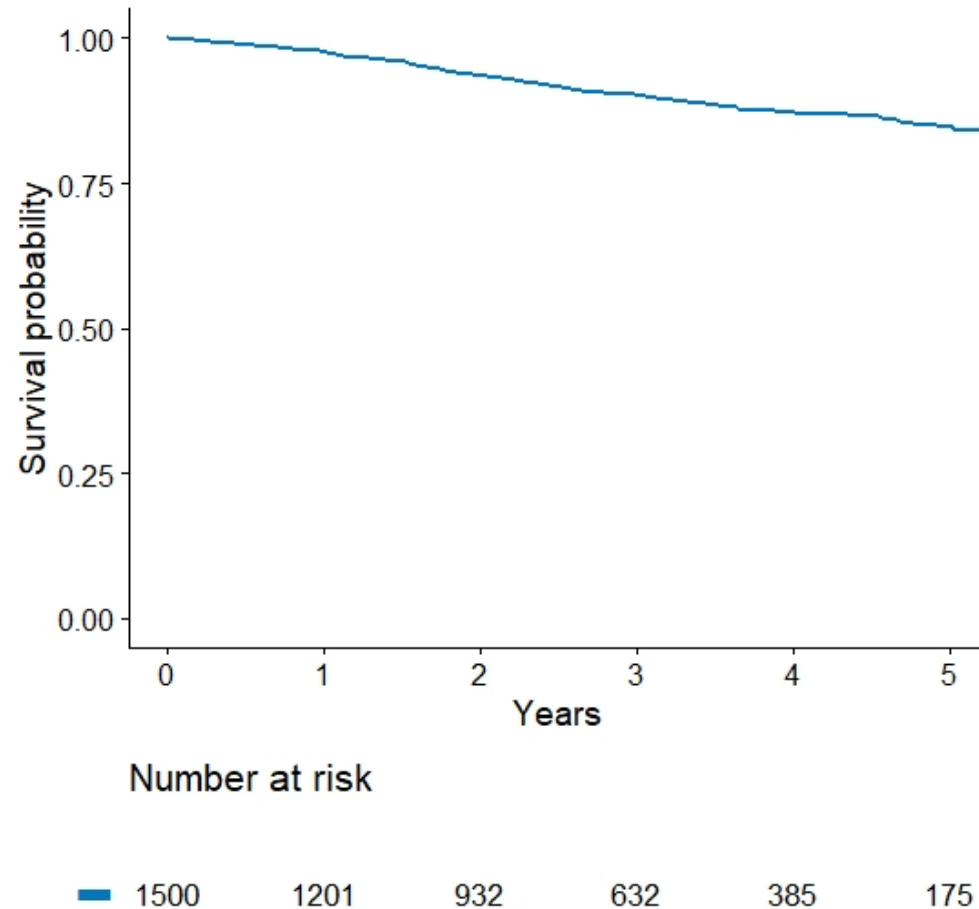
- How many patients are contributing to the estimates over time?

Tip #5: Include number at risk on K-M plots

- We see very few patients are at risk by 6 years



Tip #6: truncate K-M plot when few at risk



- Truncated x-axis from 6 to 5 years



Applying appropriate levels of precision

Characteristic	No response, N = 132	Response, N = 61	p-value
Age	46.00 (36.00, 55.00)	49.00 (43.00, 58.75)	0.091
Unknown	7	3	
Marker Level (ng/mL)	0.59 (0.20, 1.24)	0.98 (0.31, 1.53)	0.099
Unknown	6	4	
Grade			0.600
I	37 (28.0%)	22 (36.1%)	
II	47 (35.6%)	22 (36.1%)	
III	41 (31.1%)	14 (23.0%)	
IV	7 (5.3%)	3 (4.9%)	
Female	34 (25.8%)	39 (63.9%)	<0.001

- A typical Table 1 may round every continuous summary to the same number of decimal places, every percentage to the same number of decimal places, and every p-value to the same number of digits

Appropriate precision for p-values

Characteristic	No response, N = 132	Response, N = 61	p-value
Age	46.00 (36.00, 55.00)	49.00 (43.00, 58.75)	0.091
Unknown	7	3	
Marker Level (ng/mL)	0.59 (0.20, 1.24)	0.98 (0.31, 1.53)	0.10
Unknown	6	4	
Grade			0.6
I	37 (28.0%)	22 (36.1%)	
II	47 (35.6%)	22 (36.1%)	
III	41 (31.1%)	14 (23.0%)	
IV	7 (5.3%)	3 (4.9%)	
Female	34 (25.8%)	39 (63.9%)	<0.001

- A more thoughtful solution: “<0.001”, 3 significant figures for >0.001 to <0.1, 2 significant figures for 0.1-0.2, 1 significant figure for 0.21-0.9, and “>0.9”

Appropriate precision for continuous variables

Characteristic	No response, N = 132	Response, N = 61	p-value
Age	46 (36, 55)	49 (43, 59)	0.091
Unknown	7	3	
Marker Level (ng/mL)	0.59 (0.20, 1.24)	0.98 (0.31, 1.53)	0.099
Unknown	6	4	
Grade			0.600
I	37 (28.0%)	22 (36.1%)	
II	47 (35.6%)	22 (36.1%)	
III	41 (31.1%)	14 (23.0%)	
IV	7 (5.3%)	3 (4.9%)	
Female	34 (25.8%)	39 (63.9%)	<0.001

- Typically 2 significant figures (not 2 decimal places) is sufficient

Appropriate precision for percentages

Characteristic	No response, N = 132	Response, N = 61	p-value
Age	46 (36, 55)	49 (43, 59)	0.091
Unknown	7	3	
Marker Level (ng/mL)	0.59 (0.20, 1.24)	0.98 (0.31, 1.53)	0.099
Unknown	6	4	
Grade			0.600
I	37 (28%)	22 (36%)	
II	47 (36%)	22 (36%)	
III	41 (31%)	14 (23%)	
IV	7 (5.3%)	3 (4.9%)	
Female	34 (26%)	39 (64%)	<0.001

- Up to 3 significant figures is typically sufficient

Appropriate precision for ORs/HRs

Characteristic	OR	95% CI	p-value
Marker Level (ng/mL)	1.35	0.94, 1.93	0.10

- Odds ratios and hazard ratios are typically reported to two decimal places
- A single decimal place is acceptable for very large ORs/HRs, e.g. 18.2

Summary

- Focus on presenting the main study estimates of interest, typically one estimate per question/hypothesis of interest
- Always include a measure of statistical precision alongside study estimates, such as a confidence interval
- Apply appropriate levels of precision to all reported numbers

Reference

Recommendations presented here are being used by European Urology, Journal of Urology, and BJU International in statistical review of all articles:

Assel M, Sjoberg D, Elders A, Wang X, Huo D, Botchway A, Delfino K, Fan Y, Zhao Z, Koyama T, Hollenbeck B, Qin R, Zahnd W, Zabor EC, Kattan MW, Vickers AJ. Guidelines for Reporting of Statistics for Clinical Research in Urology. 2019.